



inClass- L17

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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Flow

- Announcement
- Recap
- M1C3 begins

M1C3

One variable → Many variables →
Engineering equations → Uncertainty
propagation → Approximate propagation
→ Dependence beyond correlation

M1C3 lec-1

Joint Random Variables: Describing Multiple Uncertainties

How to describe uncertainty when two variables exist simultaneously?

Single RV → Joint Distribution → Marginals →
Conditional distributions → Covariance →
Correlation → Correlation $\neq$ Independence

Message: from uncertainty in **one variable** to
uncertainty in **systems**

M1C3-Lec-1

Mon dd, 2026

Joint probability distributions

1. Earthquake response of a building

Building A is in low seismicity.

Chennai

Building B is in high seismicity.

Japan, Himalayan, California

Both have the same structural **capacity** distribution **C**.

But their earthquake **demands** **D** are different.

rv

Which building is more likely to fail?

$$Pr(\text{fail}) = Pr(D > C)$$

→ B

Dam / *D*: floods etc.
 \ *C*: open of mouth

Joint probability distributions

2. Mutual funds

Mutual funds A & B : they both lose money with probability 10%.

What is the prob. that both lose money?

^{A & B}
fully dependent: losing money in A $\rightarrow$ sure that loss in B
10%.

fully independent: $\rightarrow$ 1%.

Same individual prob $\Rightarrow$ Different Joint Probabilities

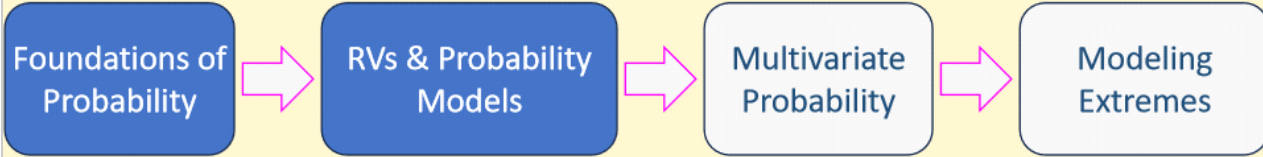
Joint Distribution of 2 or more RVs is important piece of Info.

Joint Probability Distributions

Ang and Tang (Section 3.3)

Course Flow

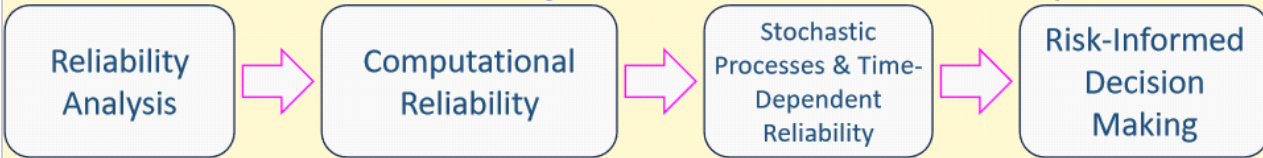
Module 1: **Probability** – Understanding Uncertainty



Module 2: **Statistics** – Learning from Data

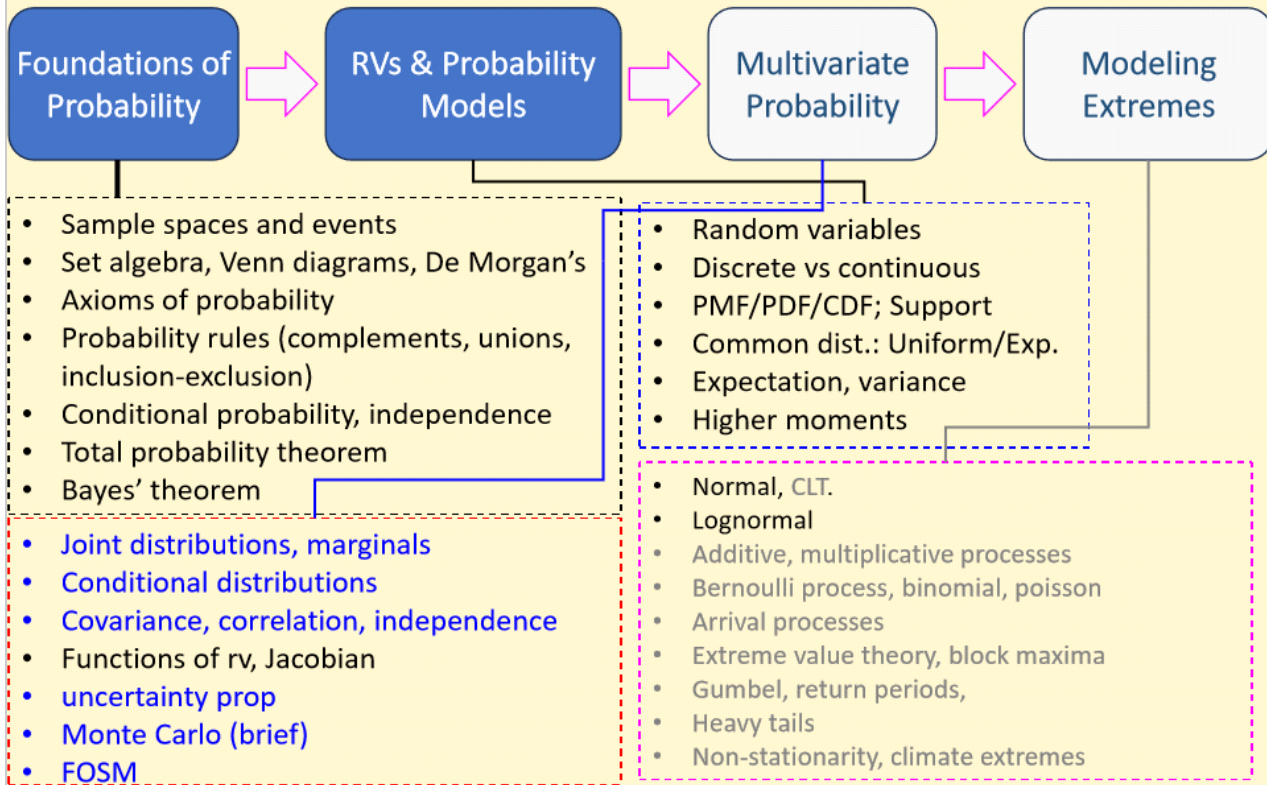


Module 3: **Reliability** – Decision under Uncertainty



Module I: Chapter 3 (M1C3)

Module 1: **Probability** – Understanding Uncertainty



Joint probability distributions

Joint CDF of X & Y .

$$F_{XY}(x, y) = \Pr(X \leq x, Y \leq y)$$

$$\rightarrow (X \leq x) \cap (Y \leq y)$$

$$F_X(x) =$$

$$\Pr(X \leq x)$$

Properties

$$\bullet F_{XY}(-\infty, -\infty) = 0$$

$$\bullet F_{XY}(\infty, \infty) = 1$$

$$F_{XY}(-\infty, y) = 0 \quad ; \quad F_{XY}(x, -\infty) = 0$$

$$F_{XY}(\infty, y) = F_Y(y) \quad ; \quad F_{XY}(x, \infty) = F_X(x)$$

$F_{XY}(x, y)$ is non-decreasing function.
non-negative

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Joint distributions

Joint PMF
2 Discrete RVs.

$$P_{XY}(x_i, y_j) = \Pr(X = x_i, Y = y_j)$$

Joint CDF

$$F_{XY}(x_i, y_j) = \sum_{x \leq x_i, y \leq y_j} P_{XY}(x, y)$$

Recall
 $F_X(x_i) = \sum_{x \leq x_i} P_X(x)$

Joint PDF
2 continuous RVs

$$f_{XY}(x, y) = \Pr(x < X \leq x + dx, y < Y \leq y + dy)$$

Joint CDF

$$F_{XY}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{XY}(u, v) du dv$$

Rel. b/w PDF & CDF

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$$

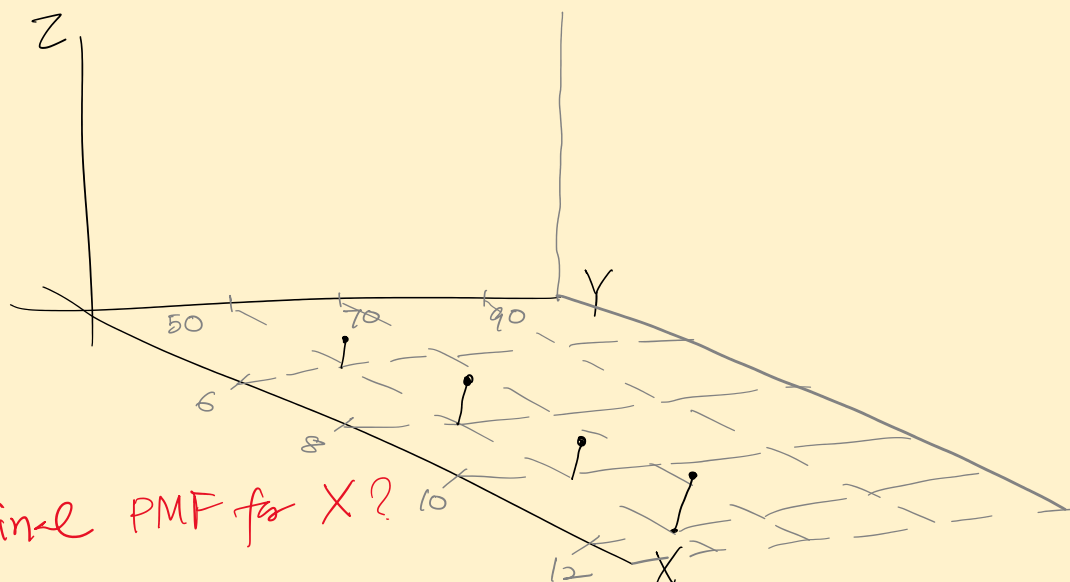
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Joint PMF

Duration and Productivity (x, y)	No. of Observations	Relative Frequency
6, 50	2	0.014
6, 70	5	0.036
6, 90	10	0.072
8, 50	5	0.036
8, 70	30	0.216
8, 90	25	0.180
10, 50	8	0.058
10, 70	25	0.180
10, 90	11	0.079
12, 50	10	0.072
12, 70	6	0.043
12, 90	2	0.014
Total = 139		

$$p_{Y|X}(90\% | 8 \text{ hr}) = \frac{p_{XY}(X=8 \text{ hr}, Y=90\%)}{p_X(X=8 \text{ hr})}$$

→ conditional (pointing to the numerator)
 → Joint (pointing to the numerator)
 → marginal (pointing to the denominator)



Marginal PMF for X?

Marginal PMF for Y?